

SAME SIZE ARTWORK
LEAFLET SIZE : 264 mm x 640 mm
FOLDING SIZE: 66 mm x 40 mm

processes account for about 35% to 40%. Biliary clearance contributes to the non-renal clearance, but the majority of non-renal clearance is thought to be due to metabolism.

Following inhalation of single and repeated once-daily doses between 62.4 mcg and 249.6 mcg glycopyrronium by healthy volunteers and patients with COPD, mean oral clearance of glycopyrronium was in the range of 17.4 L/h and 24.4 L/h indicating active tubular secretion contributes to the renal elimination of glycopyrronium.

Glycopyrronium plasma concentrations declined in a multi-phasic manner. The mean terminal elimination half-life was much longer after inhalation (33 to 52 hours) than after intravenous (8.2 hours) and oral (2.8 hours) administration.

Formoterol fumarate

Formoterol fumarate capsules were administered to 12 healthy volunteers. The mean terminal elimination half-life was approximately 10% and 7% of the dose, in intravenous and COPD patients treated for 12 weeks with 12 or 24 micrograms formoterol fumarate b.i.d., respectively. Approximately 4% of the dose was recovered in the urine as unchanged formoterol. In patients with COPD, approximately 4% of the dose was recovered in the urine as unchanged formoterol after multiple dosing of 12 and 24 micrograms. The (R,R) and (S,S) enantiomers accounted, respectively, for 40% and 60% of urinary recovery of unchanged formoterol, after single doses (12 or 24 micrograms) in healthy volunteers and after single and repeated doses in asthma patients. After a single oral dose of 3H-formoterol, 59-62% of the dose was recovered in the urine and 35-34% in the faeces. Renal clearance of formoterol is 150 mL/min.

After inhalation, plasma formoterol kinetics and urinary excretion rate data in healthy volunteers indicate a biphasic elimination, with the terminal elimination half-life of the (R,R) and (S,S) enantiomers being 13.5 and 12.3 hours, respectively. Approximately 6.44% of the dose was recovered in the urine as unchanged formoterol, with the (R,R) and (S,S) enantiomers contributing 40% and 60% respectively.

Lineation/stability

Glycopyrronium

Linear pharmacokinetics of glycopyrronium was observed following inhalation of daily doses of 31.2 mcg to 249.6 mcg. In COPD patients both systemic exposure and total urinary excretion of glycopyrronium in pharmacokinetic steady state increased dose-proportionally over the dose range of 64 to 176 mcg.

Special populations

Gender, Age, Race

Glycopyrronium

Population pharmacokinetic analysis showed no evidence of a clinically relevant effect of age (40 to 85 years) or body weight (45 to 120 kg) on systemic exposure to glycopyrronium. In addition, there was no evidence of a clinically significant ethnic/race effect (Asian, Caucasian, Chinese, Hispanic/Latino, Japanese subjects). Gender, smoking status, and baseline FEV1 have no apparent effect on mean oral or average glycopyrronium systemic exposure.

Formoterol fumarate

Elderly Patients (older than 65 years)

The pharmacokinetics of formoterol has not been studied in the elderly population. The available data from clinical trials performed in elderly patients do not suggest that the dosage should be different from other adults.

Glycopyrronium

The effects of hepatic impairment on the pharmacokinetics of glycopyrronium have not been studied. Glycopyrronium is cleared predominantly from systemic circulation by renal excretion.

Formoterol fumarate

There is no clinical reason to suggest that formoterol dosage requires adjustment in patients with hepatic impairment, however no clinical data have been generated to support its use in these groups.

Patients with renal impairment

Glycopyrronium

Formoterol fumarate was evaluated in an impact on the systemic exposure to glycopyrronium. A moderate mean increase in total systemic exposure (AUC₀₋₂₄) of about 1.4-fold was seen in subjects with mild and moderate renal impairment (estimated GFR greater than or equal to 30 mL/min/1.73m²) and up to 2.2-fold in subjects with severe renal impairment and end-stage renal disease (estimated GFR less than 30 mL/min/1.73m²).

Formoterol fumarate

There is no clinical reason to suggest that formoterol dosage requires adjustment in patients with renal impairment, however no clinical data have been generated to support its use in these groups.

Pharmacological data

Glycopyrronium

Carcinogenesis, Mutagenesis

Glycopyrronium produced no treatment-related increases in the incidence of tumors in a 2-year inhalation study in Wistar rats at inhaled doses up to 0.56 mg/kg/day (approximately 170 times the MHD in adults on an AUC basis, respectively).

A 28-week oral (gavage) study in male and female F344 rats that received glycopyrronium at doses up to 93.8 and 125.1 mg/kg/day, respectively, did not show any evidence of carcinogenicity.

Glycopyrronium tested negative in the following genotoxicity assays: *in vitro* Ames assay, *in vitro* human lymphocyte chromosomal aberration assay, and *in vivo* rat bone marrow micronucleus assay.

Animal Toxicology

Eye findings were observed in Wistar rats at inhaled doses of 0.67 mg/kg/day and higher (approximately 280 times the MHD in adults on an AUC basis) based upon findings of anterior capsule opacity, prominent anterior vitreous, and anterior capsule. No eye findings in Wistar rats were observed at an inhaled dose of 0.69 mg/kg/day (approximately 225 times the MHD in adults on an AUC basis). Eye findings were observed in beagle dogs at an inhaled dose of 0.3 mg/kg/day (approximately 100 times the MHD in adults on an AUC basis) based upon findings of corneal epithelial hyperemia and conjunctival exudate. No eye findings in beagle dogs were observed at an inhaled dose of 0.12 mg/kg/day (approximately 100 times the MHD in adults on an AUC basis).

Formoterol fumarate

Mutagenicity

Mutagenicity tests covering a broad range of experimental endpoints have been conducted. No genotoxic effects were found in any of the *in vitro* or *in vivo* tests performed.

Carcinogenicity

Two-year studies in rats and mice did not show any carcinogenic potential.

Male mice treated at very high-dose levels showed a slightly higher incidence of benign adrenal subcapsular cell tumors, which are considered to reflect alterations in the physiological aging process.

Two studies in rats, covering different dose ranges, showed an increase in mesovascular leiomyomas. These benign neoplasms are typically associated with long-term treatment of rats at high doses of 25 adrenergic drugs. Increased incidences of ovarian cysts and benign granulosa-theca cell tumors were also seen; (3 agencies are known to have effects of this type in rats which are very rarely specific to rodents). No other tumor types noted in the first study using the higher doses were within the incidences of the historical control population, and were not seen in the lower-dose experiment.

None of the tumor incidences were increased to a statistically significant extent at the lowest dose of the second study, a dose leading to a systemic exposure 10 times higher than that expected from the maximum recommended dose of formoterol.

On the basis of these findings and the absence of a mutagenic potential, it is concluded that use of formoterol at therapeutic doses does not present a carcinogenic risk.

Reproductive toxicity

Animal data have shown no teratogenic effects. Formoterol was evaluated for its effect on fertility and general reproductive performance in sexually mature male and female rats. Reproduction studies in rats revealed no impairment of fertility or effect on early embryonic development at oral doses up to 3 mg/kg (approximately 100 times the maximum recommended daily inhaled powder dose in human on a mg/m² basis).

After oral administration, formoterol was excreted in the milk of lactating rats.

PHARMACEUTICAL PARTICULARS

List of excipients:

Lactose Monohydrate & hydroxypropyl methylcellulose capsules

Incompatibilities:

Not Applicable

Shelf life:

24 months

Special precautions for storage:

Store below 30°C. Store the capsules in original package & do not remove until immediately before use.

Nature and contents of container:

Available as a carton containing 1 device, a leaflet and a blister pack of 3 x 10 capsules

Special precautions for disposal and other handling:

Not Applicable

Product Registration Holder:

Glenmark Pharmaceuticals (Malaysia) Sdn Bhd, D-31-02, Menara Satecap 1, No. 2, Jalan Kenirah, 55200 Kuala Lumpur, Malaysia

Manufacturing Site Address:

Glenmark Pharmaceuticals Limited, Village Karamnagar, Road - Nalgah Road, Tahsil Baddi, Solan (H.P.) - 173205 India

PRODUCT REGISTRATION NO.

MAJ200600742

DATE OF FIRST AUTHORIZATION/RENEWAL OF THE AUTHORIZATION

NA

DATE OF REVISION OF THE TEXT

Glenmark Airz™ F
Glycopyrronium 12.5mcg and
Formoterol Fumarate 12mcg Powder for Inhalation

Important: Do not swallow capsules. Capsules are used only with the Inhaler. Do not place a capsule in the mouthpiece of the Inhaler.

Read this Patient Information that comes with Glycopyrronium + Formoterol fumarate Powder for Inhalation before you start using it and each time you get a refill.

There may be new information.

This Patient Information does not take the place of talking to your healthcare provider about your medical condition or treatment.

Read all of this leaflet carefully before you start using this medicine because it contains important information for you.

- Keep this leaflet. You may need to read it again.
- If you have any further questions, ask your doctor, pharmacist or nurse.
- This medicine has been prescribed for you only. Do not pass it on to others. It may harm them, even if their signs of illness are the same as yours.
- If you get any side effects, talk to your doctor, pharmacist or nurse. This includes any possible side effects not listed in this leaflet.

What is in this leaflet

- What is Glycopyrronium + Formoterol fumarate Powder for Inhalation?
- Who should not use Glycopyrronium + Formoterol fumarate Powder for Inhalation?
- What should tell your healthcare provider before using Glycopyrronium + Formoterol fumarate Powder for Inhalation?
- How should you use Glycopyrronium + Formoterol fumarate Powder for Inhalation?
- What are the possible side effects of Glycopyrronium + Formoterol fumarate Powder for Inhalation?
- How should you store Glycopyrronium + Formoterol fumarate Powder for Inhalation?
- What are the ingredients in Glycopyrronium + Formoterol fumarate Powder for Inhalation?

What is Glycopyrronium + Formoterol fumarate Powder for Inhalation?

The name of product "Glycopyrronium 12.5mcg and Formoterol Fumarate 12mcg Powder for Inhalation" is referred as "Glycopyrronium + Formoterol fumarate Powder for Inhalation" in this entire leaflet.

This is a fixed-dose combination of anticholinergic Glycopyrronium 12.5mcg and long-acting beta₂-agonist (LABA) Formoterol Fumarate 12mcg powder for Inhalation.

- Anticholinergic medicines such as Glycopyrronium help the muscles around the airways in your lungs stay relaxed to prevent symptoms such as wheezing, coughing, chest tightness, and shortness of breath. This medicine relaxes the muscles around the airways.
- Formoterol fumarate is a long-acting beta₂-agonist. It makes breathing easier by relaxing muscle spasms in the air passages of the lungs. It is also used to treat breathing problems in patients with chronic obstructive pulmonary disease (COPD).
- Glycopyrronium + Formoterol fumarate Powder for Inhalation is used for maintenance treatment of COPD. COPD is a chronic lung disease that includes chronic bronchitis, emphysema, or both.
- Glycopyrronium + Formoterol fumarate Powder for Inhalation is used for long-term use and should be taken 2 times each day to improve symptoms of COPD by relieving breathing.
- Glycopyrronium + Formoterol fumarate Powder for Inhalation is not used to treat sudden symptoms of COPD. Always have a short-acting beta₂-agonist medicine (rescue inhaler) with you to treat sudden symptoms of COPD. If you do not have a rescue inhaler, contact your healthcare provider to have one prescribed for you.
- Glycopyrronium + Formoterol fumarate Powder for Inhalation should not be used in children. If you have a Glycopyrronium + Formoterol fumarate Powder for Inhalation, make sure it is effective in children younger than 12 years of age.

Who should not use Glycopyrronium + Formoterol fumarate Powder for Inhalation?

- If you are allergic to glycopyrronium, formoterol fumarate or any of the ingredients in Glycopyrronium + Formoterol fumarate Powder. Ask your healthcare provider if you are not sure. See "What are the ingredients in Glycopyrronium + Formoterol fumarate Powder?" at the end of this leaflet for a complete list of ingredients in Glycopyrronium + Formoterol fumarate Powder.
- If you have sudden symptoms of COPD.

What should tell your healthcare provider before using Glycopyrronium + Formoterol fumarate Powder for Inhalation?

Before you use using Glycopyrronium + Formoterol fumarate Powder for Inhalation, tell your healthcare provider if you:

- Have kidney problems.
- Have eye problems such as glaucoma. Glycopyrronium + Formoterol fumarate Powder for Inhalation may make your glaucoma worse.
- Have problems or bladder problems, or problems passing urine. Glycopyrronium + Formoterol fumarate Powder for Inhalation may make these problems worse.
- Have any other medical conditions.
- Take any other heart medicines.
- Have high blood pressure.
- Have an enlarged prostate gland.
- Have an overactive thyroid gland.
- Are diabetic.
- Have an arrhythmia (weak where an artery is swollen because the wall of the artery is weak).
- Have a pheochromocytoma (tumor of the adrenal gland that can affect your blood pressure).
- Are pregnant or plan to become pregnant. It is not known if Glycopyrronium + Formoterol fumarate Powder for Inhalation can harm your unborn baby.
- Are breastfeeding or plan to breastfeed. It is not known if the medicine in Glycopyrronium + Formoterol fumarate Powder for Inhalation passes into your breast milk and if it can harm your baby. You and your healthcare provider should decide if you will take Glycopyrronium + Formoterol fumarate Powder for Inhalation while breastfeeding.
- Are taking Glycopyrronium + Formoterol fumarate Powder for Inhalation or any of its ingredients, any other medicines, or food products.
- Glycopyrronium + Formoterol fumarate Powder for Inhalation contains lactose (milk sugar) and a small amount of milk proteins. It is possible that allergic medicines may happen to you if you are taking. Have nearby alert in right time if you are taking any other medicines, particularly:

- Medicines for high blood pressure.
- Beta blockers (beta₁ and beta₂ blockers).
- Heart medicines (beta₁ and beta₂ blockers).
- Heart medicines (beta₁ and beta₂ blockers).
- Medicines for depression, for example, mianserin and antidepressants (MAOIs) or tricyclics.
- Antiarrhythmics or cold remedies (including any bought from the chemist).
- Medicines called diuretics. Your doctor will know which these are.
- Symptomatic agents, which are substances that medicine used to treat asthma and nasal congestion.
- Phenothiazines, which are a group of medicines which control mental disorders such as schizophrenia, mania, psychotic conditions and anxiety.
- Anticholinergics, for example atropine or scopolamine, which are a class of medicines used to treat asthma and obstructive pulmonary disease.
- Antacids for heartburn (symptoms caused by acid reflux in the stomach).
- Antiemetics like haloperidol (used to treat nausea and vomiting).
- Anticholinergics like atropine (used to treat dry mouth and dry eyes).
- Anticholinergics like atropine (used to treat dry mouth and dry eyes).

Tell your healthcare provider about all the medicines you take, including prescription medicines, over-the-counter medicines, vitamins, and herbal supplements. Glycopyrronium + Formoterol fumarate Powder for Inhalation may affect the way other medicines work, and other medicines may affect the way Glycopyrronium + Formoterol fumarate Powder for Inhalation works. Using Glycopyrronium + Formoterol fumarate Powder for Inhalation with other medicines may cause serious side effects.

Especially tell your healthcare provider if you take anticholinergics (including anticholinergic, isotretinoin, meprobamate, atropine, glycopyrronium). Know the medicines you take. Keep a list of your medicines with you and show it to your healthcare provider and pharmacist when you get a new medicine.

Pregnancy, breast-feeding and fertility

If you are pregnant or breast-feeding, think you may be pregnant or are planning to have a baby, ask your doctor or pharmacist for advice before taking this medicine.

Driving and using machines

In some patients, Glycopyrronium + Formoterol fumarate Powder for Inhalation has been reported to cause dizziness. If you feel dizzy, do not drive or operate machinery.

Glycopyrronium + Formoterol fumarate Powder for Inhalation contains lactose

If you have been told by your doctor that you have an intolerance to some sugars, contact your doctor before taking this medicinal product.

How should you use Glycopyrronium + Formoterol fumarate Powder for Inhalation?

Read the step-by-step instructions for using Glycopyrronium + Formoterol fumarate Powder for Inhalation at the end of this Patient Information. Always take this medicine exactly as your doctor or pharmacist has told you. Check with your doctor or pharmacist if you are not sure.

- Do not use Glycopyrronium + Formoterol fumarate Powder for Inhalation unless your healthcare provider has taught you how to use the inhaler and you understand how to use it correctly.
- Use Glycopyrronium + Formoterol fumarate Powder for Inhalation exactly as your healthcare provider tells you to use it. Do not use Glycopyrronium + Formoterol fumarate Powder for Inhalation more often than prescribed for you.
- Do not swallow capsules. Only use capsules with the Inhaler.
- Use 1 capsule inhaled through the Inhaler 2 times each day (1 capsule in the morning and 1 capsule in the evening).
- Make sure that the cap has been taken, then you should open the Inhaler to check that there is no powder left in the capsule. As long as the capsule is empty, you have received 1 full dose.
- If you miss a dose of Glycopyrronium + Formoterol fumarate Powder for Inhalation, take it as soon as you remember. Take your next dose at your usual time.
- Do not use 2 capsules at 1 time.
- Do not use more than 2 capsules in a day.
- Capsules should always be taken in the original pack and only removed immediately before use.
- Always use the new Glycopyrronium + Formoterol fumarate Powder for Inhalation Inhaler that is provided with each new prescription.
- Glycopyrronium + Formoterol fumarate Powder for Inhalation does not relieve sudden symptoms of COPD. Always have a rescue inhaler medicine with you to treat sudden symptoms. If you do not have a rescue inhaler medicine, call your healthcare provider to have a rescue inhaler prescribed for you.
- Do not stop using Glycopyrronium + Formoterol fumarate Powder for Inhalation or other medicines to control or treat your COPD unless told to do so by your healthcare provider because your symptoms might get worse. Your healthcare provider will change your medicines as needed.
- Call your healthcare provider or get emergency medical care right away if your breathing problems worsen with Glycopyrronium + Formoterol fumarate Powder for Inhalation, you need to use your rescue medicine more often than usual, or your rescue inhaler medicine does not work as well for you as relieving your symptoms.

What are the possible side effects of Glycopyrronium + Formoterol fumarate Powder for Inhalation?

Use all medicines. Use medicines can cause side effects, although not every body gets them.

Glycopyrronium causes serious side effects, including:

- Sudden shortness of breath immediately after use of Glycopyrronium.** Sudden shortness of breath may be life-threatening. If you have sudden breathing problems immediately after inhaling your medicine, stop using Glycopyrronium and call your healthcare provider or go to the nearest hospital emergency room right away.
- Serious allergic reactions.** Stop using Glycopyrronium and call your healthcare provider or get emergency medical care right away if you get any of the following symptoms of a serious allergic reaction:
 - rash
 - swelling of the tongue, lips, and face
 - hives
 - difficulty breathing or swallowing
- New or worsened eye problems including acute narrow-angle glaucoma.** Acute narrow-angle glaucoma can cause permanent loss of vision if not treated. Symptoms of acute narrow-angle glaucoma may include:
 - eye pain or discomfort
 - nausea or vomiting
 - blurred vision
 - seeing halos or bright spots around lights
 - red eyes

If you have these symptoms, stop taking Glycopyrronium and call your healthcare provider right away before using another dose.

- New or worsened urinary retention.** People who use Glycopyrronium may develop new or worse urinary retention. Urinary retention can be caused by a blockage in your bladder. Urinary retention can also happen to men who have a larger than normal prostate.
- Symptoms of urinary retention may include:**
 - difficulty urinating
 - painful urination
 - urinating frequently
 - urination in a weak stream or drops
- If you have these symptoms, stop taking Glycopyrronium and call your healthcare provider right away before taking another dose.

Common side effects of Glycopyrronium include upper respiratory tract infection, sore throat and runny nose.

Formoterol may cause serious side effects, including:

Some side effects can be serious.

- Stop using formoterol and tell your doctor straight away if you notice:
 - Bronchospasm when you start coughing and difficulty in breathing. This may affect up to 1 in 100 people.
 - A allergic reaction when you start taking this medicine. This may include hives, rash, or difficulty breathing or facial swelling. This may affect up to 1 in 100 people.
 - If you feel you have muscle weakness, muscle cramps or an abnormal heart rhythm (these could mean you have a low blood potassium level). This may affect up to 1 in 100 people.
 - If you feel you have an irregular heart beat or a fast heart beat. This may affect up to 1 in 100 people.
 - If you feel a crushing chest pain (symptoms of angina pectoris). This may affect up to 1 in 10,000 people.

The side effects listed below have also been reported.

Common (may affect up to 1 in 100 people)

Headache, tremor, palpitations.

Uncommon (may affect up to 1 in 100 people)

Agitation, anxiety, feeling nervous, dizziness, loss of appetite, drowsiness, dry mouth, muscle cramps, muscle pain, worsening of asthma.

Rare (may affect up to 1 in 1,000 people)

Nausea.

Very rare side effects (may affect up to 1 in 10,000 people)

Distorted sense of taste, swelling of hands, ankles or feet, variations in blood pressure.

Known side effects (may affect up to 1 in 10,000 people)

Distorted sense of taste, swelling of hands, ankles or feet, variations in blood pressure.

Not known (frequency cannot be estimated from the available data)

Headache and dizziness (symptoms of high blood pressure), cough and rash.

Some of these effects may wear off as you get used to the medicine.

If any of the symptoms become troublesome, or if you notice anything else not mentioned here, please go and see your doctor. He/she may want to give you a different medicine.

Reporting of side effects

If you get any side effects, talk to your doctor or pharmacist. This includes any possible side effects not listed in this leaflet. By reporting side effects you can provide valuable information on the safety of this medicine.

How should you store Glycopyrronium + Formoterol fumarate Powder for Inhalation?

Store below 30°C. Store the capsules in original package. Do not remove until immediately before use.

What are the ingredients in Glycopyrronium + Formoterol fumarate Powder for Inhalation?

Lactose Monohydrate & hydroxypropyl methylcellulose capsules

Instructions for Use
Glycopyrronium + Formoterol fumarate Powder for Inhalation

How to use your inhaler:

- Remove the protection cover from Inhaler P. Hold the base firmly and mouth piece in the direction given on the device.
- Insertion: Insert the capsule into the chamber. Place the capsule properly into the slot as shown. Rotate the mouth piece back in position to close the device.
- Puncture the capsule. Hold the lower between your thumb and index finger. Puncture the capsule firmly in order to evenly puncture the capsule, make sure that you push both the buttons inward, simultaneously.
- Turn your head to the side and breathe out (exhale) as much as you can. Your lungs should be as empty as possible right before you breathe in the medicine.
- Put your mouth on the Inhaler P's mouthpiece. Grip the mouthpiece between your teeth and close your lips firmly around the Inhaler P, so that no air or medicine can escape out. Then your head upwards.
- Using only your mouth, breathe in deep and fast. Continue breathing in, until your lungs are completely full of air. Hold the Inhaler P in "breath-activated" position. This means that breathing in fast and deep makes the right dose of medicine come out of the Inhaler P. Then the medicine goes directly into your lungs.
- Take your mouth off the Inhaler P's mouthpiece and hold your breath for 10 seconds or as long as possible. This will hold the medicine inside your airways and change and help you breathe easier. Slowly breathe out through your nose. Check that the capsule is empty and the total dose is consumed. If any dose remains in the capsule, repeat steps until you have taken the number of doses, as prescribed by your physician. Ensure that you give an interval of 1 min between two doses.
- Remove the empty capsule and put the protective cover back. This helps to keep dirt and moisture (wetness) out of the Inhaler P. Store in safe and dry place. Moisture or dirt may cause Inhaler P to stop up or stop working correctly.
- After taking the dose, wipe your mouth and brush your teeth.

Cleaning:

Clean your Inhaler P with a brush or wipe with a clean, dry, lint-free cloth to remove any powder residue. While cleaning the capsule chamber, make sure you push the buttons inward to ensure complete cleaning. Do not use water for cleaning.

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ICONGRAPHICS CODE:

PANTONE SHADE



PANTONE
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PROCESS C

**Supersedes
Artwork Code:**

Product Registration Holder:
Glenmark Pharmaceuticals (Malaysia) Sdn Bhd
D-31-02, Menara Satecap 1,
No. 2, Jalan Kenirah,
55200 Kuala Lumpur, Malaysia

Manufactured by:
glenmark
Pharmaceuticals Ltd.
Glenmark Pharmaceuticals Limited
B/2, Mahalaksh Chambers
22, Brabham Road
Mumbai - 400 026, INDIA
At: Unit Village Karamnagar, Baddi Nalgah Road,
Tahsil Baddi, District Solan (H.P.) 173205, India.
TM Trade Mark

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